

1. Abstract Class vs. Interface

What is the main difference between an abstract class and an interface in PHP?

Abstract Class

Partial Abstraction: An abstract class can define both abstract methods (with no implementation) and concrete methods (with implementation).

Inheritance: A class can only inherit from one abstract class directly (but can implement multiple interfaces).

Data Members: Can contain properties and variables.

Access Modifiers: Can use access specifiers (public, private, protected) for methods and properties.

Use Case: Provides a blueprint for subclasses with some pre-defined behavior and enforces the implementation of specific methods.

Interface

Full Abstraction: An interface only defines abstract methods, outlining the functionality required without any implementation details.

Inheritance: A class can implement multiple interfaces, promoting loose coupling and flexibility.

No Data Members: Interfaces cannot contain properties or variables.

Access Specifiers: Methods are implicitly public in interfaces.

Use Case: Defines a contract for a specific behavior that any class can adhere to.